

ISA CERTIFIED PERSONAL TRAINER COURSE

WEEK 6

- CHAPTER 10
(MUSCULAR TRAINING: ASSESSMENTS)



CHAPTER 10
MUSCULAR TRAINING:
ASSESSMENTS

CONTENT COVERED



Selection and administration of appropriate assessments based on ACE Mover Method™



Evaluation of results of assessments and application to Muscular Training



When and why the repeat of Muscular Training assessments



Explanation of Muscular Training assessments with clients

ACE INTEGRATED FITNESS TRAINING MODEL

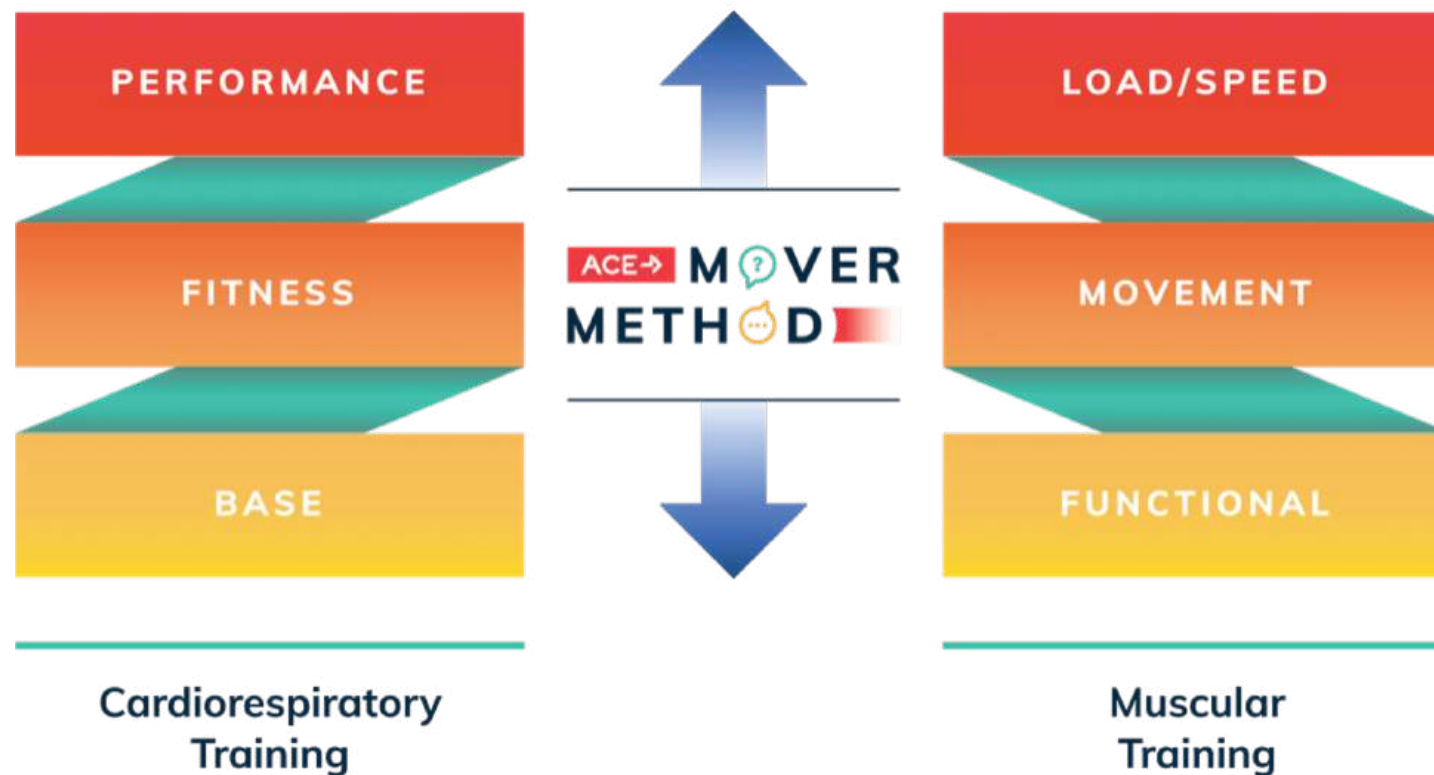


Figure 2-2: ACE IFT Model

FUNCTIONAL ASSESSMENTS

- Functional Training
 - Focus: Establishing or re-establishing postural stability and kinetic chain mobility
 - How: By improving joint function through

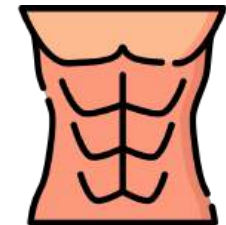
Improved muscular endurance



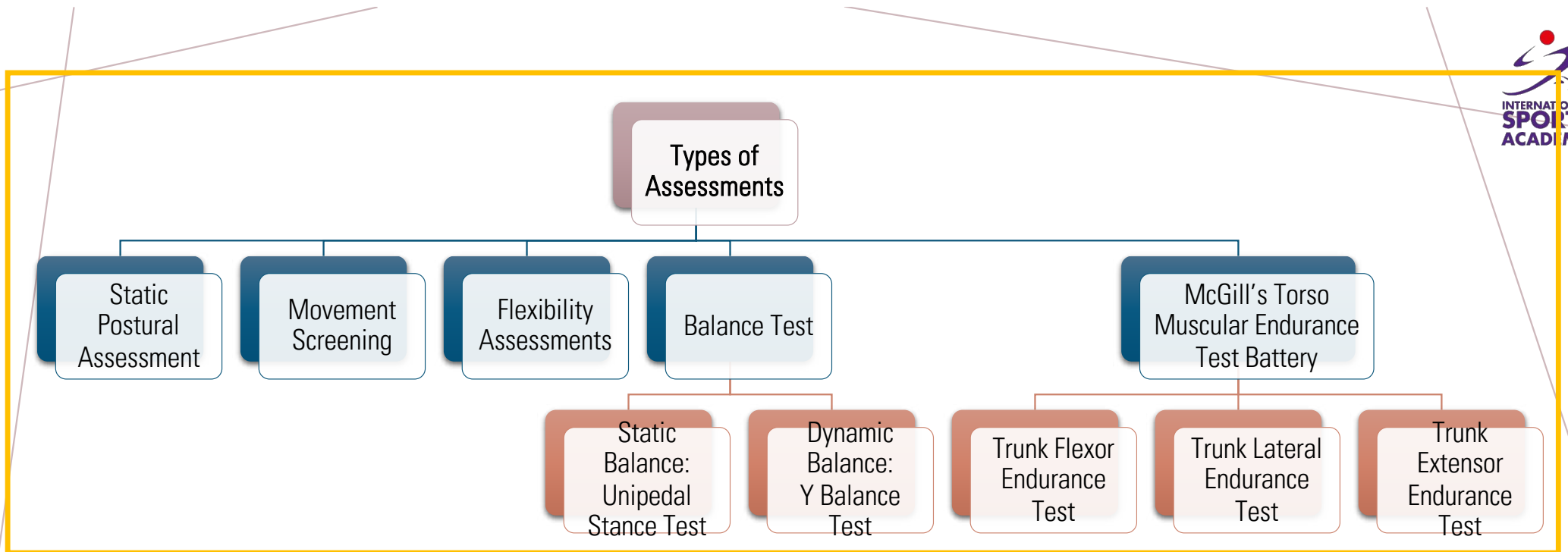
Static Balance Dynamic balance



Flexibility



Core function



STATIC POSTURAL ASSESSMENT

- Static Posture
 - Alignment of body's segments
- Assessment may offer insights into:
 - Muscle imbalance
 - Altered neural action of the muscles

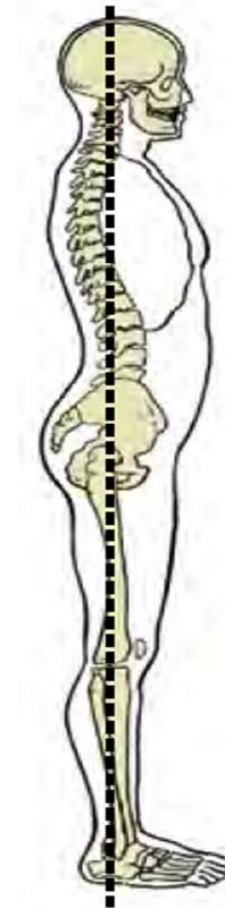


Figure 10-2:

Natural spine alignment with slight anterior (lordotic) curves at the neck and low back and a posterior (kyphotic) curve in the thoracic region

STATIC POSTURAL ASSESSMENT

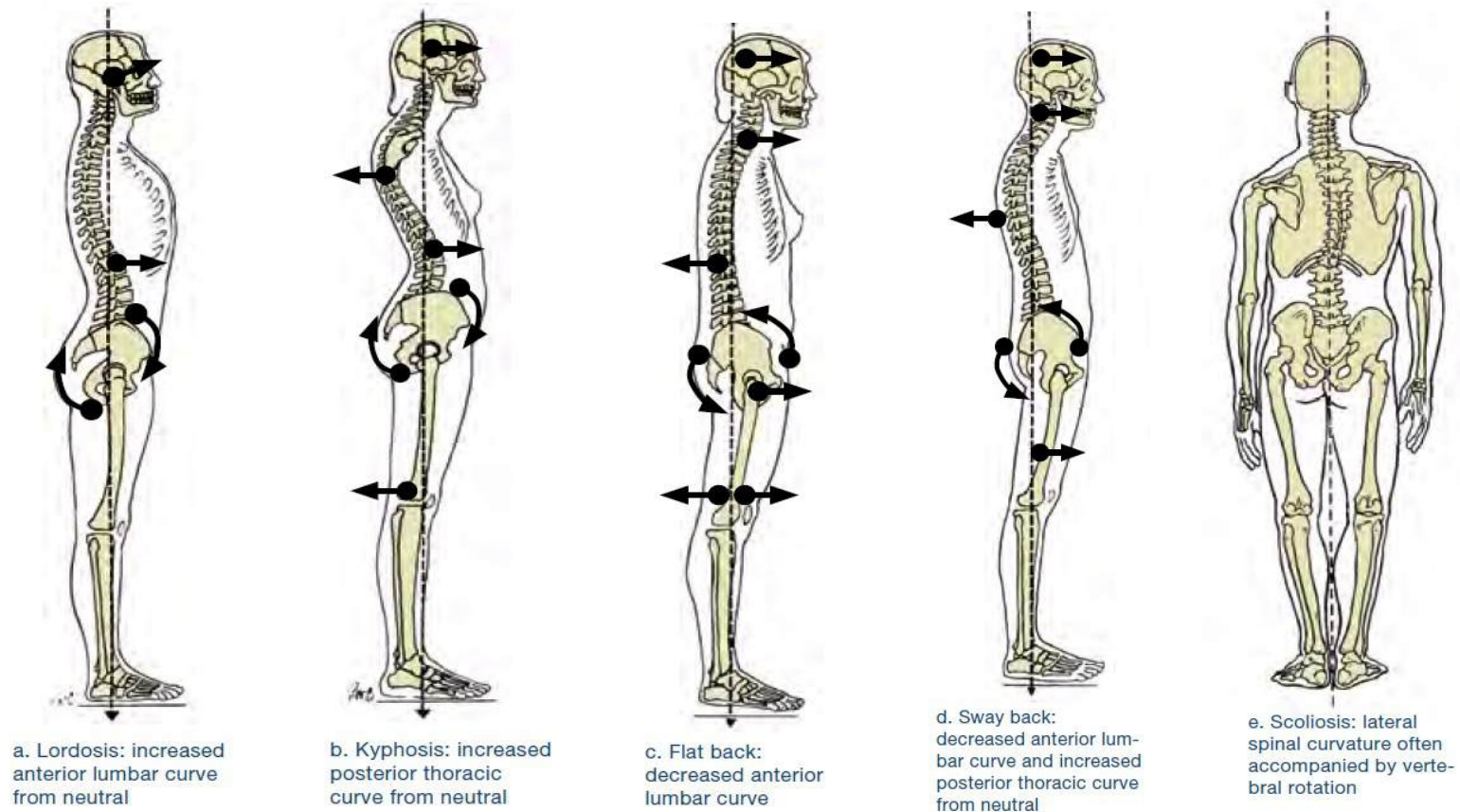


Figure 10-3: Postural Deviations

Posture	Facilitated/Hypertronic (shortened)	Inhibited (lengthened)
Lordosis	Hip flexors	Hip extensors
	Lumbar extensors	External obliques
		Rectus abdominis

Table 10-1: Muscle imbalances associated with Lordosis posture

Posture	Facilitated/Hypertronic (shortened)	Inhibited (lengthened)
Kyphosis	Anterior chest/shoulders	Upper-back extensors
	Latissimus dorsi	Scapular stabilisers
	Neck extensors	Neck flexors

Table 10-2: Muscle imbalances associated with Kyphosis posture

Posture	Facilitated/Hypertronic (shortened)	Inhibited (lengthened)
Flat-back	Rectus abdominis	Iliacus/psoas major
	Upper-back extensors	Lumbar extensors
	Neck extensors	Neck flexors
	Ankle plantar flexors	

Table 10-3: Muscle imbalances associated with Flat-back posture

Posture	Facilitated/Hypertronic (shortened)	Inhibited (lengthened)
Sway-back	Hamstrings	Iliacus/psoas major
	Upper-fibers of posterior obliques	Rectus femoris
	Lumbar extensors	External oblique
	Neck extensors	Upper-back extensors
		Neck flexors

Table 10-4: Muscle imbalances associated with Sway-back posture

MUSCLE BALANCE AND POSTURAL DEVIATIONS

Correctible Factors

- Repetitive movements (muscular pattern overload)
- Awkward positions and movements (habitually poor posture)
- Side Dominance
- Lack of joint stability
- Imbalanced strength-training programs

Non-correctible Factors

- Congenital conditions (e.g., scoliosis)
- Some pathologies (e.g., rheumatoid arthritis)
- Structural deviations (e.g., tibial or femoral torsion, or femoral anteversion)
- Certain types of trauma (e.g., surgery, injury, or amputation)

RIGHT ANGLE RULE

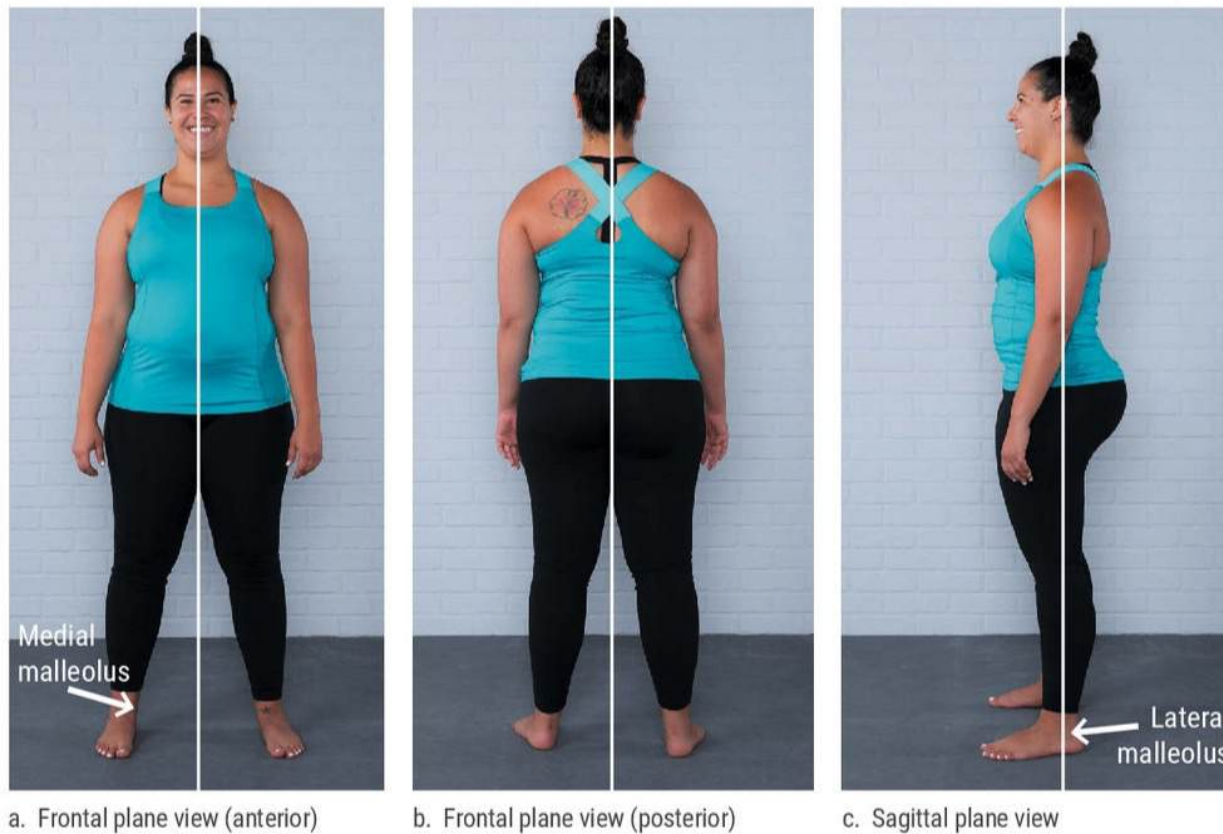
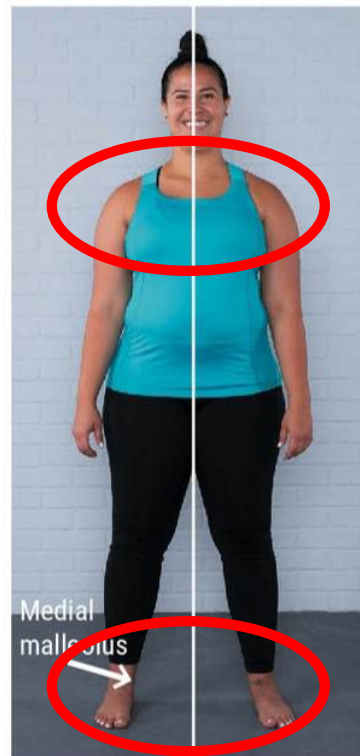


Figure 10-4: The right angle rule (frontal and sagittal plane views)

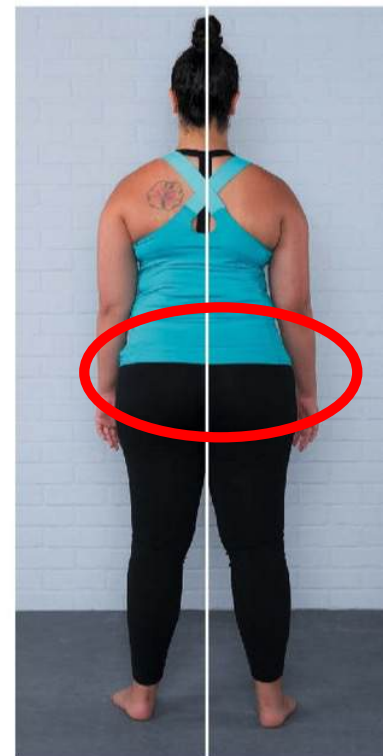
- When assessing posture, observe the client's stance following the right angle rule
- Right-angle model implies the body is in a good position at different planes

COMMON STATIC POSTURE DEVIATIONS

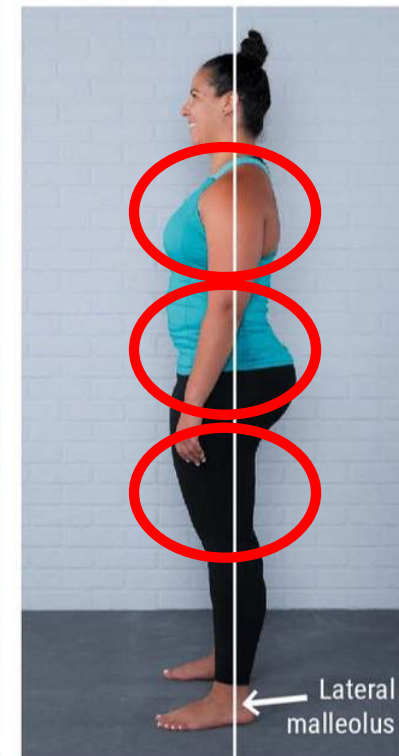
- How the human body represents itself:
 1. Ankle (and subtalar joint)
 2. Knee
 3. Hip
 4. Shoulder
 5. Head
- Look in all three planes to note any static asymmetries



a. Frontal plane view (anterior)



b. Frontal plane view (posterior)



c. Sagittal plane view

VIDEO HERE

- <https://www.youtube.com/watch?v=cxOQR-yccsQ&t=132s>
- Short video on how to assess static postural deviations (5 deviations on the following slides)
- Item required: long dowel

DEVIATION 1: SUBTALAR PRONATION/SUPINATION

- Both feet should face forward in parallel or with slight (12 to 18 degrees) external rotation.
- Toes should be aligned in the same direction as the feet.
- Any excessive pronation (arch flattening) or supination (high arches) at the subtalar joint should be noted.

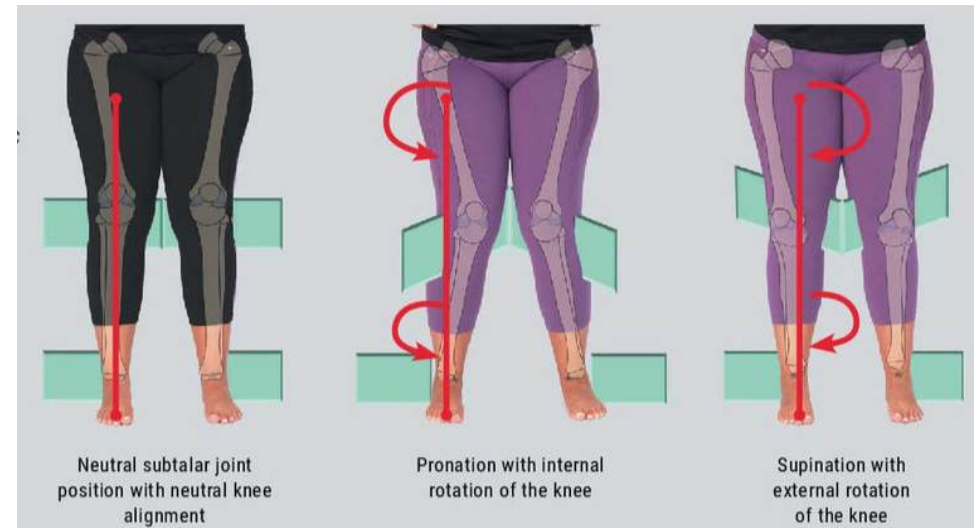
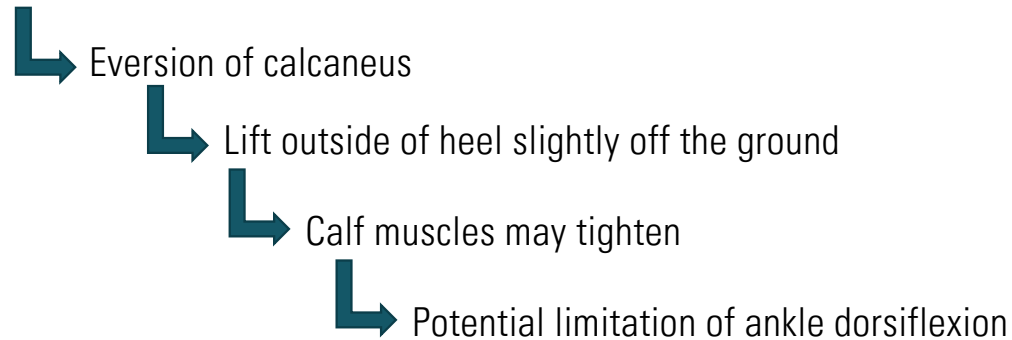


Figure 10-5: Foot pronation and supination and the effects up the kinetic chain

DEVIATION 1: SUBTALAR PRONATION/SUPINATION

Pronation of subtalar joint



Subtalar Joint Pronation/Supination and the Effect on the Feet, Tibia, and Femur				
Subtalar Joint Movement	Foot Movement	Tibial (Knee) Movement	Femoral Movement	Plane of View
Subtalar joint pronation	Eversion	Internal rotation	Internal rotation	View from front
Subtalar joint supination	Inversion	External rotation	External rotation	View from front

Table 10-5: Subtalar joint pronation/supination and the effect on Feet, Tibia and Femur

DEVIATION 2: HIP ADDUCTION

- Hip adduction: lateral tilt of the pelvis that elevates one hip higher than the other (hip hiking)
- May be evident in people with leg-length discrepancies
- Progressively lengthens and weakens right hip abductors

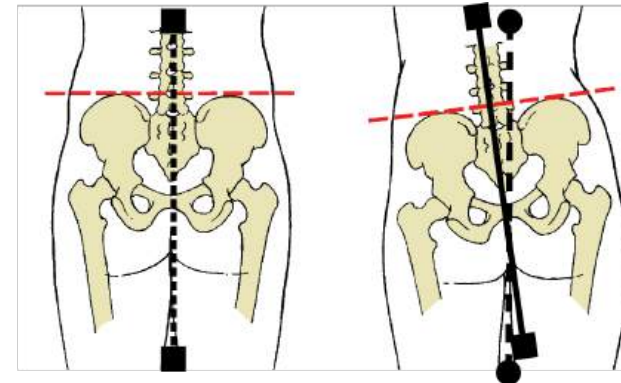


Figure 10-6: Normal hip position versus right hip adduction (posterior view)

Observation	Position	Alignment	Frontal Plane
Right hip adduction	Elevated (vs. left side)	Hips usually shifted right	Posterior view
Left hip adduction	Elevated (vs. right side)	Hips usually shifted left	Posterior view

Table 10-6: Hip Adduction

DEVIATION 3: PELVIC TILTING

- Anterior pelvic tilt: rotates superior, anterior portion of the pelvis forward and downward
- Posterior pelvic tilt: rotates superior posterior portion of the pelvis backward and downward

	Anterior Tilt	Posterior Tilt
Rotation	ASIS tilts downward and forward	ASIS tilts upward and backward
Muscles suspected to be tight	Hip flexors, erector spinae	Rectus abdominis, hamstrings
Muscles suspected to be lengthened	Hamstrings, rectus abdominis	Hip flexors, erector spinae
Plane of view	Sagittal	Sagittal

Note: ASIS = Anterior superior iliac spine

Table 10-7: Pelvic tilt

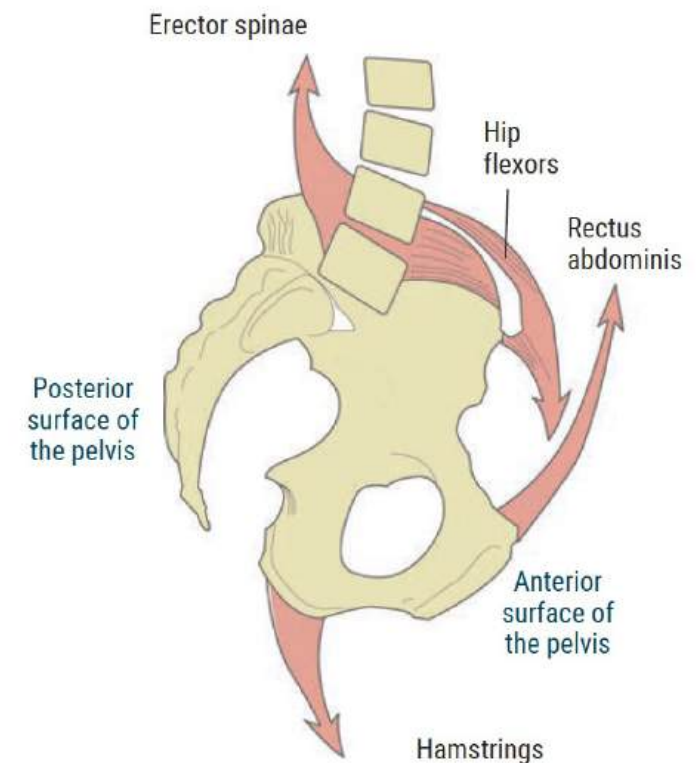
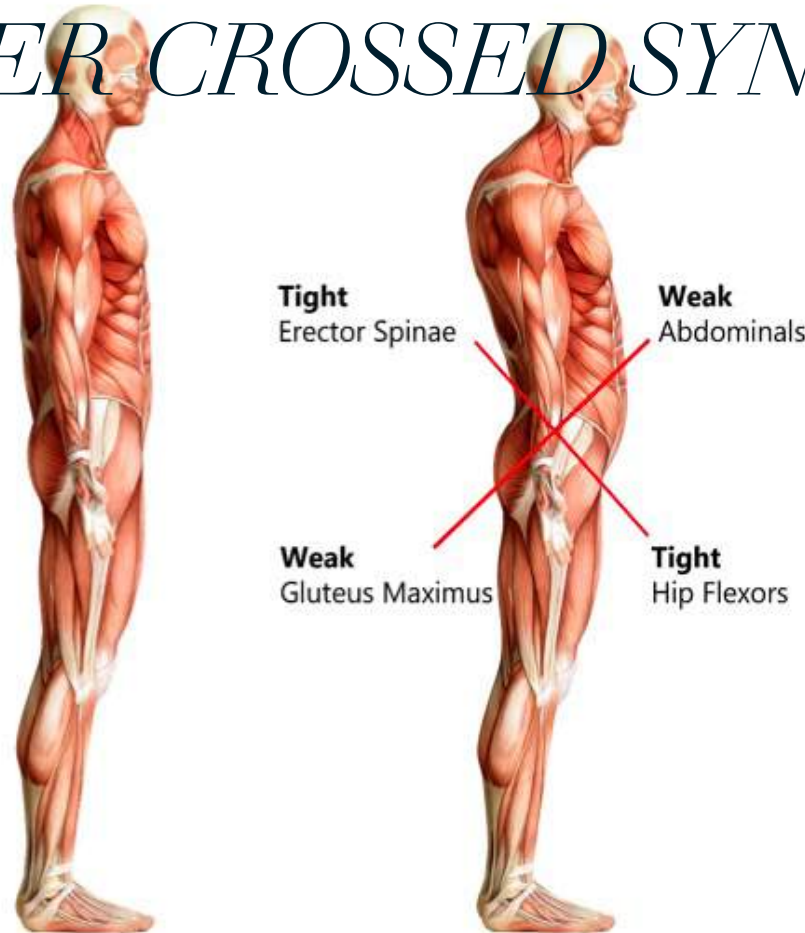


Figure 10-8: Muscular control of the pelvis

LOWER-CROSSED SYNDROME



ustrated in Figure 7-11, an anterior pelvic tilt rotates

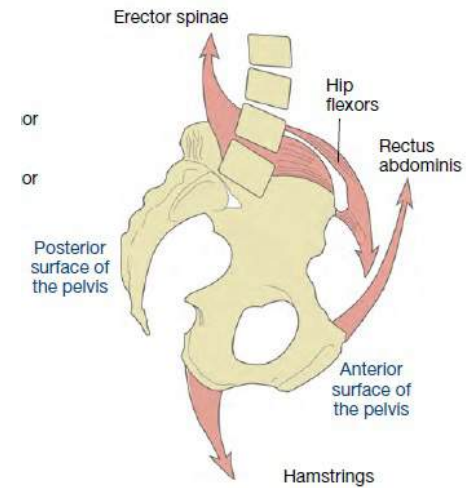


Figure 7-10
Muscular control of the pelvis by the abdominals and hip flexors (iliopsoas and rectus femoris) anteriorly and the spinal extensors (erector spinae) and hamstrings posteriorly

DEVIATION 4: SHOULDER POSITION AND THE THORACIC SPINE

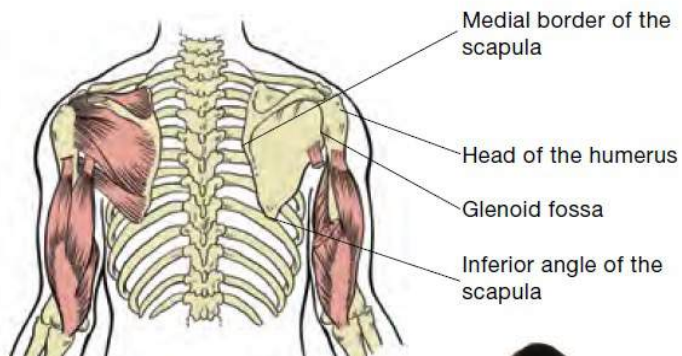


Figure 10-14: The normal position of the scapulae

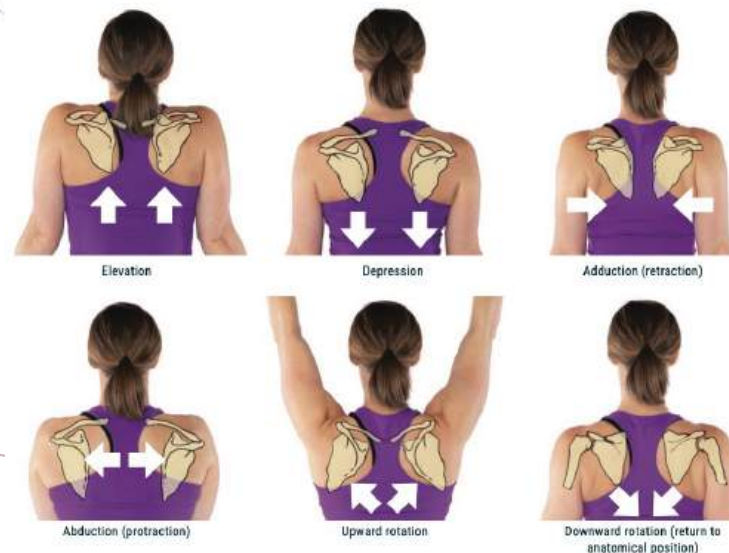


Figure 10-13:
Scapular movement

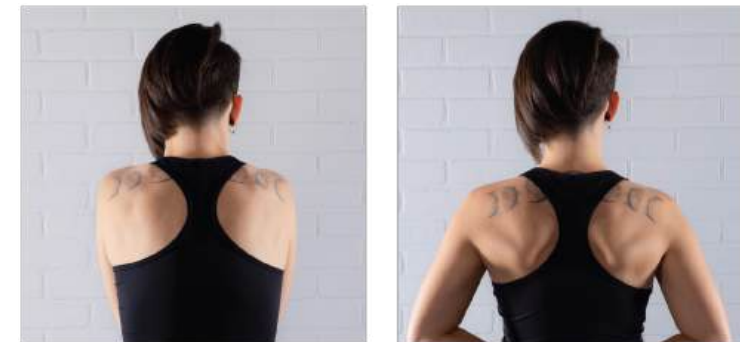


Figure 10-15: Scapular protraction and winging:
Posterior view



Figure 10-16:
Scapular
protraction:
Anterior view

Scapular protraction:
Anterior view

DEVIATION 4: SHOULDER POSITION AND THE THORACIC SPINE

Observation	Muscles Suspected to Be Tight	Plane of View
Shoulders not level	Upper trapezius, levator scapula, rhomboids on the elevated side	Frontal
Asymmetry to midline	Lateral trunk flexors (flexed side)	Frontal
Protracted (forward, rounded)	Serratus anterior*, anterior scapulo-humeral muscles†, upper trapezius	Sagittal
Medially rotated humerus	Pectoralis major and latissimus dorsi (shoulder adductors), subscapularis	Frontal
Kyphosis and depressed chest	Shoulder adductors, pectoralis minor, rectus abdominis, internal oblique	Sagittal

*Serratus anterior is usually tight with scapular protraction and is usually lengthened with scapular winging.

†A group of seven muscles (coracobrachialis, deltoids, four rotator cuff muscles, and teres major) that connect the humerus to the scapula and function to stabilize the glenohumeral joint.

Table 10-8: Shoulder position

DEVIATION 5: HEAD POSITION

- Good posture: earlobe should align approximately over the acromion process
- Forward head posture: head shifts forward, earlobe appears significantly forward of the acromioclavicular (AC) joint

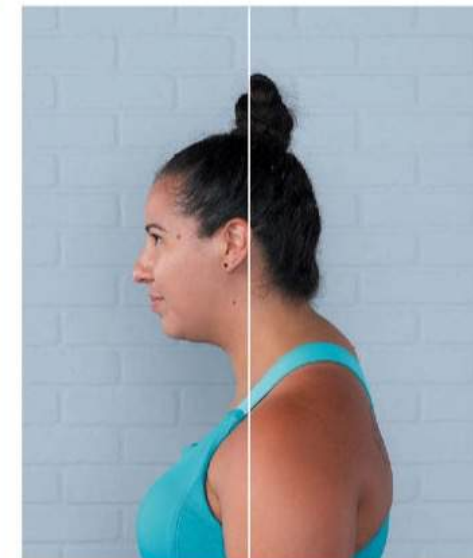
Head Position

Table 10-9: Head position

Observation	Muscles Suspected to Be Tight	Plane of View
Forward-head position	Cervical spine extensors, upper trapezius, levator scapulae	Sagittal



Postural alignment



Forward-head position

Figure 10-17:
Alignment of the
acromioclavicular
joint with the ear

POSTURAL ASSESSMENT CHECKLIST

FRONTAL VIEW	
☐	Overall body symmetry: symmetrical alignment of the left and right hemispheres
☐	Ankle position: observe for pronation and supination
☐	Foot position: observe for inversion and eversion
☐	Knees: rotation and height discrepancies
☐	Hip adduction and shifting: observe for shifting to a side as witnessed by the position of the pubis in relation to the plumb line
☐	Alignment of the iliac crests
☐	Alignment of the torso: position of the umbilicus and sternum in relation to the plumb line
☐	Alignment of the shoulders
☐	Arm spacing: observe the space to the sides of the torso
☐	Hand position: observe the position relative to the torso
☐	Head position: alignment of the ears, nose, eyes, and chin

POSTERIOR VIEW	
☐	Overall body symmetry: symmetrical alignment of the left and right hemispheres
☐	Alignment of the spine: vertical alignment of the spinous processes (may require forward bending)
☐	Alignment of the scapulae: inferior angle of scapulae and presence of winged scapulae
☐	Alignment of the shoulders
☐	Head: alignment of the ears

SIDE VIEW	
☐	Overall body symmetry: symmetrical alignment of load-bearing joint landmarks with the line of gravity
☐	Knees: flexion or extension
☐	Pelvic alignment for tilting: relationship of ASIS to PSIS
☐	Spinal curves: observe for thoracic kyphosis, lumbar lordosis, or flat-back position
☐	Shoulder position: forward rounding (protraction) of the scapulae
☐	Head position: neutral cervical curvature (versus forward position) and level (position above the clavicle)

Note: ASIS = Anterior superior iliac spine; PSIS = Posterior superior iliac spine

Figure 10-18: Postural assessment checklist

STATIC BALANCE: UNIPEDAL STANCE TEST

- Objective: Assess static balance by standing with a reduced base of support while removing visual sensory information
- Termination criteria:
 - Client opens eyes during an eyes-closed trial
 - Dominant weight-bearing foot moves in order to maintain balance
 - Nondominant foot touches the floor or moves away from the dominant weight-bearing limb
 - Client achieves the maximum 45-second duration of the test



Figure 10-21:
Unipedal
Stance
Test

Age Category (years)	Eyes Open (seconds)		Eyes Closed (seconds)	
	Female	Male	Female	Male
18–39	45.1	44.4	13.1	16.9
40–49	42.1	41.6	13.5	12.0
50–59	40.9	41.5	7.9	8.6
60–69	30.4	33.8	3.6	5.1
70–79	16.7	25.9	3.7	2.6
80–99	10.6	8.7	2.1	1.8

Reprinted with permission from Gibson, A.L., Wagner, D.R., & Heyward, V.H. (2019). *Advanced Fitness Assessment and Exercise Prescription* (8th ed.). Champaign, Ill.: Human Kinetics, page 354.

Table 10-10: Normative Data for the Unipedal Stance Test

VIDEO

- <https://www.youtube.com/watch?v=PzaeZjsVs5Q>
- This video demonstrates how to perform the unipedal stance test
- Note the starting position, ending position, and termination criteria
- Items required: Nil

DYNAMIC BALANCE: Y BALANCE TEST

- Objective: assess dynamic balance
- Asymmetries between legs during this test are related to injury risk and may indicate the need for guided injury-prevention strategies.
- Appropriate for athletic populations and clients with sport-related goals
- Not recommended for older adult clients/those with balance impairments

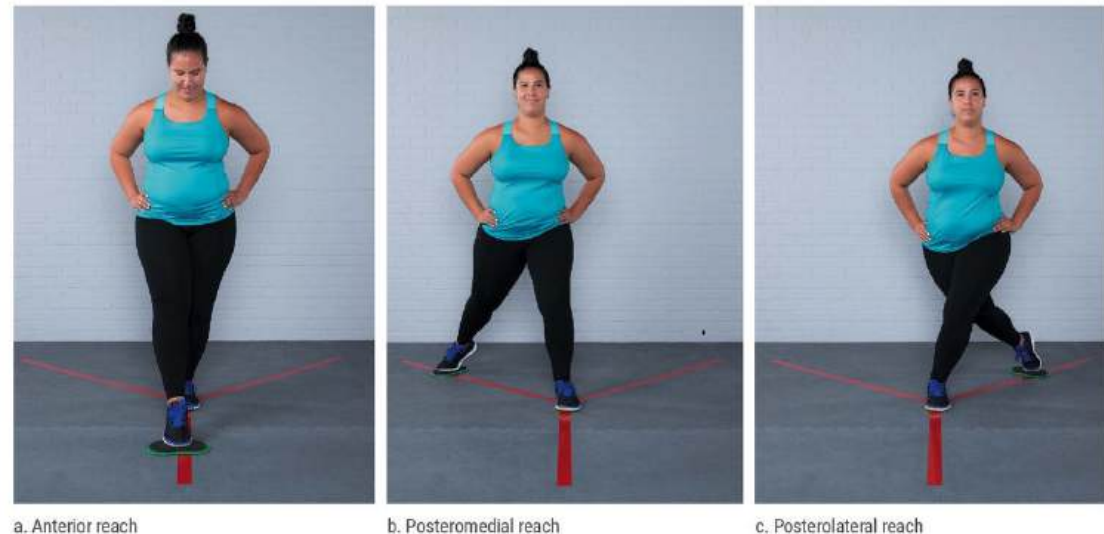


Figure 10-22: Three directions of the Y balance test

Direction	Right Foot Stance (Left Foot Reach)	Left Foot Stance (Right Foot Reach)	Difference
Anterior			
Posteromedial			
Posterolateral			

Table 10-11: Scoring sheet
for Y-balance test

VIDEO

- <https://www.youtube.com/watch?v=qFVuS-P2FrI>
- This video demonstrates how to perform the Y balance test
- Note the starting position, how the movement of the legs are during the test, and take away (as indicated in the previous slide)
- Items required: tape with the measured distance (refer to video)

MCGILL'S TORSO MUSCULAR ENDURANCE TEST BATTERY

Trunk Flexor
Endurance Test

Trunk Lateral
Endurance Test

Trunk Extensor
Endurance Test

TRUNK FLEXOR ENDURANCE TEST

- First in the battery of the three tests
- Objective: Assess muscular endurance of the anterior deep core muscles
- Contraindications
 - Low-back pain
 - Recent back surgery
 - Acute low-back flare up



Figure 10-25: Trunk flexor endurance test

TRUNK LATERAL ENDURANCE TEST

- Objective: Assess muscular endurance of the lateral core muscles
- Contraindications
 - Shoulder pain or weakness
 - Low-back pain
 - Recent back surgery
 - Acute low-back flare-up



Figure 10-27: Trunk Lateral Endurance Test

Figure 10-28: Trunk Lateral Endurance Test: Modified hook-lying position



TRUNK EXTENSOR ENDURANCE TEST

- Objective: Assess muscular endurance of the torso extensor muscles
- Contraindications
 - Client with major strength deficiencies
 - Client with high body mass
 - Low-back pain, recent back surgery, acute low-back flare-up



a. Starting position

Figure 10-30: Trunk Extensor Endurance Test



b. Test position

VIDEO

- https://www.youtube.com/watch?v=A_9mhL-_04c
- This video demonstrates how to perform the McGill's Torso Test
- Note the positions to adopt during each of the test, and the contraindications
- Items required: flat board, massage bed

MCGILL'S TORSO MUSCULAR ENDURANCE TEST BATTERY

Trunk flexor endurance test	
Time to completion: _____	
Trunk lateral endurance test	
Right side time to completion: _____ Left side time to completion: _____	
Trunk extensor endurance test	
Time to completion: _____	
Ratio of Comparison	Criteria for Good Relationship Between Muscles
Flexion:extension	Ratio less than 1.0
Right-side bridge:left-side bridge	Scores should be no greater than 0.05 from a balanced score of 1.0
Side bridge (each side):extension	Ratio less than 0.75
Flexion:extension ratio: _____ Rating: ☐ Good ☐ Poor	
Right-side bridge:left-side bridge ratio: _____ Rating: ☐ Good ☐ Poor	
Side-bridge (each side):extension ratio: _____ Rating: ☐ Good ☐ Poor	

- Goal: To create ratios consistent with McGill's recommendations

Figure 10-32: McGill's torso muscular endurance test battery – record sheet

FLEXIBILITY AND MUSCLE-LENGTH TESTING

- Objective: Assess the flexibility of specific muscle groups where tightness or limitations to movement are suspected

Joint and Movement	ROM (°)	Joint and Movement	ROM (°)
Shoulder/Scapulae		Thoraco-lumbar Spine	
Flexion	150–180	Lumbar flexion	40–45
Extension	50–60	Thoracic flexion	30–40
Abduction	180	Lumbar extension	30–40
Internal/medial rotation	70–80	Thoracic extension	20–30
External/lateral rotation	90	Lumbar rotation	10–15
Shoulder horizontal adduction	90*	Thoracic rotation	35
Shoulder horizontal abduction	30–40*	Lumbar lateral flexion	20
		Thoracic lateral flexion	20–25
Elbow		Hip	
Flexion	145	Flexion	100–120
Extension	0	Extension	10–30
		Abduction	40–45
		Adduction	20–30
		Internal/medial rotation	35–45
		External/lateral rotation	45–60

Joint and Movement	ROM (°)	Joint and Movement	ROM (°)
Radio-ulnar		Knee	
Pronation	90	Flexion	125–145
Supination	90	Extension	0–10
Wrist		Ankle	
Flexion	80	Dorsiflexion	20
Extension	70	Plantar flexion	45–50
Radial deviation	20		
Ulnar deviation	45		
Cervical Spine		Subtalar	
Flexion	45–50	Inversion	30–35
Extension	45–75	Eversion	15–20
Lateral flexion	45		
Rotation	65–75		

*Zero point (0 degrees) is with the arms positioned in frontal-plane abduction at shoulder height.

Source: Kendall, F.P. et al. (2005). *Muscles Testing and Function with Posture and Pain* (5th ed.). Baltimore, Md.: Lippincott Williams & Wilkins.

Table 10-12: Average range of motion for Healthy Adults

THOMAS TEST

- Objective: Evaluate the length of the muscles involved in hip flexion (i.e., hip flexors and rectus femoris)
- This test should not be conducted on clients suffering from low-back pain without medical clearance



a. Starting position



b. Test position

Figure 10-37: Thomas Test for Hip Flexor Length

VIDEO

- <https://www.youtube.com/watch?v=leu-t3vwbgI>
- This video demonstrates how to perform the Thomas Test
- Note the starting positions, ending position, and the contraindications
- Go through the pointers highlighted in the table in the next slide as the person in the video demonstrates the movement
- Items required: massage bed

THOMAS TEST

Movement/Limitation	Suspected Muscle Tightness
With the back and sacrum flat: ↑ The back of the lowered thigh does not touch the table ↑ The knee does not flex to 80 degrees	Primary hip flexor muscles
With the back and sacrum flat: ↑ The back of the lowered thigh does not touch the table ↑ The knee does flex to 80 degrees	The iliopsoas, which is preventing the hip from rotating posteriorly and inhibiting the thigh from being able to touch the table
With the back and sacrum flat: ↑ The back of the lowered thigh does touch the table ↑ The knee does not flex to 80 degrees	The rectus femoris, which does not allow the knee to bend

Table 10-13: Interpretation of the Thomas Test

PASSIVE STRAIGHT-LEG RAISE

- Objective: Assess length of hamstrings
- Important to note the degree of movement attained from the table or mat before the spine compresses the hand under the low back or the opposite leg begins to show visible signs of lifting off the table or mat

Figure 10-38: Passive straight leg-raise: PT's hand position



Figure 10-39: Passive straight leg-raise: Assessment position

VIDEO

- <https://www.youtube.com/watch?v=cqC4T73xIZw>
- This video demonstrates how to perform the PSL
- Note the starting positions, ending position, and the contraindications
- Go through the pointers highlighted in the table in the next slide as the person in the video demonstrates the movement
- Items required: massage bed

PASSIVE STRAIGHT-LEG RAISE

Movement/Limitation	Hamstrings Length
The raised leg achieves ≥ 80 degrees of movement before the pelvis rotates posteriorly.	Normal hamstrings length
The raised leg achieves < 80 degrees of movement before the pelvis rotates posteriorly or there are any visible signs in the opposite leg lifting off the mat or table.	Tight hamstrings

Source: Kendall, F.P. et al. (2005). *Muscles Testing and Function with Posture and Pain* (5th ed.). Baltimore, Md.: Lippincott Williams & Wilkins.

Table 10-14: Interpretation of the Passive Straight-leg raise

SHOULDER MOBILITY TEST – FLEXION AND EXTENSION

- Objective: Assess degree of shoulder flexion and extension
- Flexion:
 - Maintain extended elbows and neutral wrist position
 - Avoid arching in the low back during movement
 - Avoid depression of the rib cage, which may pull the shoulders off the mat



Figure 10-40: Shoulder Flexion Assessment

SHOULDER MOBILITY TEST – FLEXION AND EXTENSION

- Extension:
 - Small amount of extension in the thoracic spine is acceptable during the movement
 - Avoid arching in the low back or any rotation of torso during movement
 - Avoid any attempts to lift chest or head off the mat during the movement



Figure 10-41: Shoulder
Extension Assessment

VIDEO

- <https://www.youtube.com/watch?v=GL6kBBMtWkY&t=216s>
- Video starts from 2:23 to 3:08 for shoulder flexion and extension test
- Note the starting and ending position, and the caution points as highlighted in the previous slide
- Go through the pointers highlighted in the table in the next slide as the person in the video demonstrates the movement
- Items required: yoga mat

SHOULDER MOBILITY TEST – FLEXION AND EXTENSION

Movement/Limitation—Flexion	Shoulder Mobility
Ability to flex the shoulders to 170–180 degrees (hands touching/nearly touching floor)	Good shoulder mobility
Inability to flex the shoulders to 170 degrees or discrepancies between the limbs	Potential tightness in the <u>pectoralis major</u> and minor, latissimus dorsi, teres major, rhomboids, and subscapularis ▶ Tightness in the <u>latissimus dorsi</u> will force the low back to arch. ▶ Tightness of the <u>pectoralis minor</u> may tilt the scapulae forward (anterior tilt) and prevent the arms from touching the floor. ▶ Tight abdominals may depress the rib cage, tilting the scapulae forward (anterior tilt), and prevent the arms from touching the floor. ▶ Thoracic kyphosis may round the thoracic spine and prevent the arms from touching the floor.

Movement/Limitation—Extension	Shoulder Mobility
Ability to extend the shoulders to 50–60 degrees off the floor	Good shoulder mobility
Inability to extend the shoulders to 50 degrees or discrepancies between the limbs	Potential tightness in <u>pectoralis major</u>, abdominals, <u>subscapularis</u>, certain shoulder flexors (<u>anterior deltoid</u>), coracobrachialis, and biceps brachii ▶ Tightness in the <u>abdominals</u> may prevent normal extension of the thoracic spine and rib cage. ▶ Tightness in the <u>biceps brachii</u> may prevent adequate shoulder extension with an extended elbow (but may permit extension with a bent elbow).

Table 10-15: Interpretation of the Shoulder Flexion and Extension Assessment

MOVEMENT ASSESSMENTS

- Bend-and-lift assessment: squat pattern
- Single-leg assessment: step-up
- Push assessment: shoulder push stabilisation
- Pull assessment: standing row
- Rotation assessment: thoracic spine mobility

BEND-AND-LIFT ASSESSMENT: SQUAT PATTERN

- Objective: Assess symmetrical lower-extremity mobility and stability, and trunk mobility and stability during a bend-and-lift movement



Figure 10-42: Bend-and-lift assessment (squat pattern): Anterior view



Figure 10-42: Bend-and-lift assessment (squat pattern): Side view

VIDEO

- <https://www.youtube.com/watch?v=ID1KMP2AKfc>
- Video highlights the key points for performing the bend and lift screen
- Note the starting and ending position
- Go through the pointers highlighted in the table in the next slide as the person in the video demonstrates the movement
- Items required: NII

BEND-AND-LIFT ASSESSMENT: SQUAT PATTERN

Table 10-16: Bend-and-lift Assessment: Squat Pattern

View	Location	Compensation	Key Suspected Compensations: Overactive (Tight)	Key Suspected Compensations: Underactive (Lengthened)	
☐	Anterior	Feet	Lack of foot stability: Ankles collapse inward/feet turn outward	Soleus, lateral gastrocnemius, peroneals	Medial gastrocnemius, gracilis, sartorius, tibialis group
☐	Anterior	Knees	Move inward	Hip adductors, tensor fascia latae	Gluteus medius and maximus
☐	Anterior	Torso	Lateral shift to a side	Side dominance and muscle imbalance due to potential lack of stability in the lower extremity during joint loading	
☐	Side	Feet	Unable to keep heels in contact with the floor	Plantar flexors	None
☐	Side	Hip and knee	Initiation of movement	Movement initiated at knees may indicate quadriceps and hip flexor dominance, as well as insufficient activation of the gluteus group	
☐	Side	Hip and knee	Unable to achieve tops of thighs parallel to the floor	Poor mechanics, lack of dorsiflexion due to tight plantar flexors (which normally allow the tibia to move forward)	
		Contact behind knee	Hamstrings contact back of calves	Muscle weakness and poor mechanics, resulting in an inability to stabilize and control the lowering phase	
☐	Side	Lumbar and thoracic spine	Back excessively arches (i.e., lumbar dominance)	Hip flexors, back extensors, latissimus dorsi	Core, rectus abdominis, gluteal group, hamstrings
			Back rounds forward	Latissimus dorsi, teres major, pectoralis major and minor	Upper back extensors
☐	Side	Head	Downward	Increased hip and trunk flexion	
			Upward	Compression and tightness in the cervical extensor region	

SINGLE-LEG ASSESSMENT: STEP-UP

- Objective: Assess symmetrical lower-extremity mobility and stability and trunk mobility and stability during a single-leg (step-up) movement



Figure 10-45: Step-up: Anterior view

Figure 10-46: Step-up: Side view

VIDEO

- <https://www.youtube.com/watch?v=WCFCdxzFBa4>
- Do note that the video shows a step up exercise rather than a screen. So recording of the video would require to mimic the movement in the textbook (figure on previous slide)
- Note the starting and ending position
- Go through the pointers highlighted in the table in the next slide as the person in the video demonstrates the movement
- Items required: box

SINGLE-LEG ASSESSMENT: STEP-UP

View		Joint Location	Compensation	Key Suspected Compensation: Overactive (Tight)	Key Suspected Compensation: Underactive (Tight)
☐	Anterior	Feet	Lack of foot stability: Ankles collapse inward/feet turn outward	Soleus, lateral gastrocnemius, peroneals	Medial gastrocnemius, gracilis, sartorius, tibialis group, gluteus medius and maximum - inability to control internal rotation
☐	Anterior	Knees	Move inward	Hip adductors, tensor fascia latae	Gluteus medius and maximus
☐	Anterior	Hips	Hip adduction * > 2 inches (5.1cm)	Hip adductors, tensor fascia latae	Gluteus medius and maximus
			Stance-leg rotation (inward)	Stance-leg or raised-leg internal rotators	Stance-leg or raised-leg external rotators
☐	Anterior	Torso	Lateral tilt, forward lean, rotation	Lack of core stability	
☐	Anterior	Raised-leg	Lack of ankle dorsiflexion	Ankle plantar flexors	Ankle dorsiflexors
			Limb deviates from sagittal plane	Raised-leg hip extensors	Raised-leg hip flexors
			Hiking the raise hip	Stance-leg hip flexors - limiting posterior hip rotation	
☐	Side	Pelvis and low back	Anterior tilt with forward torso lean	Stance-leg hip flexors	Rectus abdominis and hip extensors
			Posterior tilt with hunched-over torso	Rectus abdominis and hip extensors	Stance-leg hip flexors

Table 10-17: Single-leg assessment: Step Up

PUSH ASSESSMENT: SHOULDER PUSH STABILIZATION

- Objective: Assess stabilization of the scapulothoracic joint and core control during closed-kinetic-chain pushing movements



Figure 10-47: Push
assessment: Shoulder push
stabilization

VIDEO

- <https://www.youtube.com/watch?v=tMsQ-Nr2q7w>
- Video highlights the movement during the shoulder push stabilization screen
- Note the starting and ending position
- Go through the pointers highlighted in the table in the next slide as the person in the video demonstrates the movement
- Items required: yoga mat

PUSH ASSESSMENT: SHOULDER PUSH STABILIZATION

View		Joint Location	Compensation	Key Suspected Compensations
☐	Side	Scapulothoracic	Exhibits “winging” during the push-up movement	Inability of the parascapular muscles (i.e., serratus anterior, trapezius, levator scapula, rhomboids) to stabilize the scapulae against the rib cage. Can also be due to a flat thoracic spine.
☐	Side	Trunk	Hyperextension or “collapsing” of the low back	Lack of core, abdominal, and low-back strength, resulting in instability

Table 10-18: Shoulder push stabilisation

PULL ASSESSMENT: STANDING ROW

- Objective: Assess movement efficiency and potential muscle imbalances during pulling movements



Figure 10-48:
Pull
Assessment:
Standing
Row

VIDEO

- <https://www.youtube.com/watch?v=qZYJPMQ7-qc>
- Focus on 0:17 to 0:32 of the video for the cable pull movement
- Note the starting and ending position
- Go through the pointers highlighted in the table in the next slide as the person in the video demonstrates the movement
- Items required: pulley system

PULL ASSESSMENT: STANDING ROW

View	Location	Compensation	Key Suspected Compensations: Overactive (Tight)	Key Suspected Compensations: Underactive (Lengthened)
Side	Lumbar spine	Hyperextension	Hip flexors, back extensors	Core, rectus abdominis, gluteal group, hamstrings
Posterior	Scapulothoracic	Elevation	Upper trapezius, levator scapulae, rhomboid major and minor	Mid and lower trapezius
Side	Head	Migrates forward (protraction)	Cervical spine extensors, upper trapezius, levator scapulae	Cervical spine flexors
Posterior	Scapulothoracic	Abduction (protraction)	Serratus anterior, anterior scapulothoracic muscles, upper trapezius	Rhomboid major and minor, middle trapezius

Table 10-19: Pull Assessment: Standing row

ROTATION ASSESSMENT: THORACIC SPINE MOBILITY

- Objective: Assess bilateral mobility of the thoracic spine
- Lumbar spine rotation is considered insignificant, as it only offers approximately 15 degrees of rotation

Figure 10-49: Rotation Assessment: Thoracic spine mobility



a. Starting position



b. Assessment position

VIDEO

- <https://www.youtube.com/watch?v=D9p04QPvoPQ>
- Focus on video from 0:41 to end for thoracic mobility movement screen
- Note the starting and ending position
- Go through the pointers highlighted in the table in the next slide as the person in the video demonstrates the movement
- Items required: long dowel

ROTATION ASSESSMENT: THORACIC SPINE MOBILITY

View		Location	Compensation	Possible Biomechanical Problems
☐	Anterior or posterior	Trunk	None if trunk rotation achieves 45 degrees in each direction	
☐	Anterior or posterior	Trunk	Bilateral discrepancy (assuming no existing congenital issues in the spine)	Side-dominance Differences in paraspinal development Torso rotation, perhaps associated with some hip rotation Note: Lack of thoracic mobility will negatively impact glenohumeral mobility

Table 10-20: Rotation Assessment: Thoracic Spine Mobility

LOAD/SPEED ASSESSMENTS

Muscular-endurance assessments

- Push-up assessment
- Body-weight squat assessment

Muscular-strength assessment

- 1-RM bench press assessment
- 1-RM squat assessment
- Submaximal strength assessment

Power assessment

- Vertical jump assessment

Speed, agility, and quickness assessment

- T-test

MUSCULAR ENDURANCE ASSESSMENTS

- Evaluate the ability of a specific muscle group, or groups, to perform repeated or sustained contractions to sufficiently invoke muscular fatigue
- Considerations:
 - Ensure proper form and posture is maintained with each repetition
 - Ensure there is no low-back pain or other orthopedic issues
 - Any pain merits immediate termination of the test and referral to a more qualified professional



PUSH-UP ASSESSMENT

- Objective: Measure upper-body endurance
 - The client completes as many repetitions as possible before reaching fatigue
 - Women should perform the modified push-up due to variations in strength between men and women



a. Standard push-up starting position



b. Modified push-up starting position



a. Standard push-up "up" position



b. Modified push-up "up" position

Figure 10-52 & 53: Push Up Assessment:

VIDEO

- <https://www.youtube.com/watch?v=7dXB7ZdOhf4>
- Focus on the movement of a push up from beginning to 2:43 of the video
- Note the starting and ending position
- Go through the pointers highlighted in the previous slide
- Items required: yoga mat, block

PUSH-UP ASSESSMENT

Category	Age (years)											
	15–19		20–29		30–39		40–49		50–59		60–69	
Sex	M	W	M	W	M	W	M	W	M	W	M	W
Excellent	≥39	≥33	≥36	≥30	≥30	≥27	≥25	≥24	≥21	≥21	≥18	≥17
Very good	29–38	25–32	29–35	21–29	22–29	20–26	17–24	15–23	13–20	11–20	11–17	12–16
Good	23–28	18–24	22–28	15–20	17–21	13–19	13–16	11–14	10–12	7–10	8–10	5–11
Fair	18–22	12–17	17–21	10–14	12–16	8–12	10–12	5–10	7–9	2–6	5–7	2–4
Needs improvement	≤17	≤11	≤16	≤9	≤11	≤7	≤9	≤4	≤6	≤1	≤4	≤1

Note: M = Men; W = Women

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Table 10-21: Converting the push up score (number) to a Health benefit rating

BODY-WEIGHT SQUAT

- Objective: evaluate muscular endurance of the lower extremity
- Only suitable for those who demonstrate proper squat form



Figure 10-54: Body-weight squat assessment

VIDEO

- <https://www.youtube.com/watch?v=4CR4jFBumb8>
- The video show the client performing a squat from 1:20 to 2:00
- Note the starting and ending position
- Go through the pointers highlighted in the previous slide
- Items required: Nil

BODY-WEIGHT SQUAT

Males	18-25	26-35	36-45	46-55	56-65	65+
Excellent	>49	>45	>41	>35	>31	>28
Good	44-49	40-45	35-41	29-35	25-31	22-28
Above average	39-43	35-39	30-34	25-38	21-24	19-21
Average	35-38	31-34	27-29	22-24	17-20	15-18
Below average	31-34	29-30	23-26	18-21	13-16	11-14
Poor	25-30	22-28	17-22	13-17	9-12	7-10
Very poor	<25	<22	<17	<9	<9	<7
Females	18-25	26-35	36-45	46-55	56-65	65+
Excellent	>43	>39	>33	>27	>24	>23
Good	37-43	33-39	27-33	22-27	18-24	17-23
Above average	33-36	29-32	23-26	18-21	13-17	14-16
Average	29-32	25-28	19-22	14-17	10-12	11-13
Below average	25-28	21-24	15-18	10-13	7-9	5-10
Poor	18-24	13-20	7-14	5-9	3-6	2-4
Very Poor	<18	<13	<7	<5	<3	<2

Table 10-22: Norms for body-weight squats

MUSCULAR STRENGTH ASSESSMENTS

- Strength as an important component of physical fitness and is dependent on a variety of factors
- Absolute strength – the greatest amount of weight that can be lifted one time
 - Known as a one-repetition maximum (1-RM)
- Relative strength – the maximum force a person can exert in relation to his or her body weight
 - $\text{Relative strength} = \text{Absolute strength} / \text{Body weight}$

1-RM BENCH-PRESS ASSESSMENT

- Objective: Measure upper-extremity strength
- Only suitable for individuals who:
 - Can demonstrate proper bench press form
 - Do not have a history of shoulder problems



Starting and ending position



Mid-range position

Figure 10-55: Bench press assessment

VIDEO

- <https://www.acefitness.org/education-and-resources/lifestyle/exercise-library/5/chest-press/>
- The website shows how to perform a bench press. For the purpose of the video, just need to show how to perform a bench press and how to spot during the movement. Execution of the 1RM protocol is not required for the purpose of the video
- Note the starting and ending position
- Go through the pointers highlighted in the previous slide
- Items required: bench, rack, barbell, weights

1-RM BENCH-PRESS ASSESSMENT

One-repetition Maximum Bench Press (Bench Press Weight Ratio = Weight Pushed / Body Weight)							
		Age					
	%	< 20	20-29	30-39	40-49	50-59	60+
Superior	95	1.76	1.63	1.35	1.20	1.05	0.94
Excellent	80	1.34	1.32	1.12	1.00	0.90	0.82
Good	60	1.19	1.14	0.98	0.88	0.79	0.72
Fair	40	1.06	0.99	0.88	0.80	0.71	0.66
Poor	20	0.89	0.88	0.78	0.72	0.63	0.57
Very Poor	5	0.76	0.72	0.65	0.59	0.53	0.49

Table 10-23: Upper body strength (Men)

One-repetition Maximum Bench Press (Bench Press Weight Ratio = Weight Pushed / Body Weight)							
		Age					
	%	< 20	20-29	30-39	40-49	50-59	60+
Superior	95	0.88	1.01	0.82	0.77	0.68	0.72
Excellent	80	0.77	0.80	0.70	0.62	0.55	0.54
Good	60	0.65	0.70	0.60	0.54	0.48	0.47
Fair	40	0.58	0.59	0.53	0.50	0.44	0.43
Poor	20	0.53	0.51	0.47	0.43	0.39	0.38
Very Poor	5	0.41	0.44	0.39	0.35	0.31	0.26

Table 10-24: Upper body strength (Women)

1-RM SQUAT ASSESSMENT

- Objective: Measure lower-extremity strength
- Unsupported and functional movement
- Only suitable for individuals who:
 - Can demonstrate proper form when performing a squat
 - Do not have low-back or knee pain



Figure 10-59: Squat Assessment

VIDEO

- <https://www.acefitness.org/education-and-resources/lifestyle/exercise-library/11/back-squat/>
- The website shows how to perform a bench press. For the purpose of the video, just need to show how to perform a squat and how to spot during the movement. Execution of the 1RM protocol is not required for the purpose of the video
- Note the starting and ending position
- Go through the pointers highlighted in the previous slide
- Items required: rack, barbell, weights

SPOTTING

- Spotting exercises/movement
 - Over the head, in front of shoulders, on the back, or moving over the face
 - When clients require assistance to finish a repetition
- Spot as close to the dumbbell as possible for dumbbell exercises
- Spotters to ensure a stable base of support and maintain good posture via neutral spine when spotting heavier weights



Figure 10-57:
Proper example
of spotting
overhead press



SUBMAXIMAL STRENGTH ASSESSMENTS

- Objective: Measure upper- and lower-extremity strength using submaximal loads
- Appropriate for inexperienced exercisers or individuals with health considerations
- Use various exercises (that do not exceed 10 repetitions) to assess strength

Repetitions	% 1-RM
1	100
2	95
3	93
4	90
5	87
6	85
7	83
8	80
9	77
10	75
11	70
12	67
15	65

Table 10-25:
1-RM
repetition
table

Note: 1-RM = One-repetition maximum

SUBMAXIMAL STRENGTH ASSESSMENTS

Joint	Movements	Muscles	Ratio
Shoulder	Flexion:Extension	Anterior deltoids:Trapezius, posterior deltoids	2:3
Shoulder	Internal rotation: External rotation	Subscapularis:Supraspinatus, infraspinatus, teres minor	3:2
Elbow	Flexion:Extension	Biceps:Triceps	1:1
Lumbar spine	Flexion:Extension	Iliopsoas, abdominals:Erector spinae	1:1
Hip	Flexion:Extension	Iliopsoas, rectus abdominis, tensor fascia latae:Erector spinae, gluteus maximus, hamstrings	1:1
Knee	Flexion:Extension	Hamstrings:Quadriceps	2:3
Ankle	Plantar flexion:Dorsiflexion	Gastrocnemius:Tibialis anterior	3:1
Ankle	Inversion:Eversion	Tibialis anterior:Peroneals	1:1

Table 10-26: Appropriate Strength Ratios

POWER ASSESSMENT

- Human power is defined as the rate at which mechanical work is performed under a defined set of conditions
- Sport or activity specific

Power Equations
$\text{Power} = \text{Force} \times \text{Velocity}$
or
$\text{Power} = \text{Work} / \text{Time}$
Where:
$\text{Force} = \text{Mass} \times \text{Acceleration}$
$\text{Velocity} = \text{Distance} / \text{Time}$
$\text{Work} = \text{Force} \times \text{Distance}$

POWER ASSESSMENTS

- **Contraindications:**

- Special populations in general
- Clients with orthopedic limitations
- Clients recovering from an injury



VERTICAL JUMP ASSESSMENT

- Objective: Measure standing vertical jump height

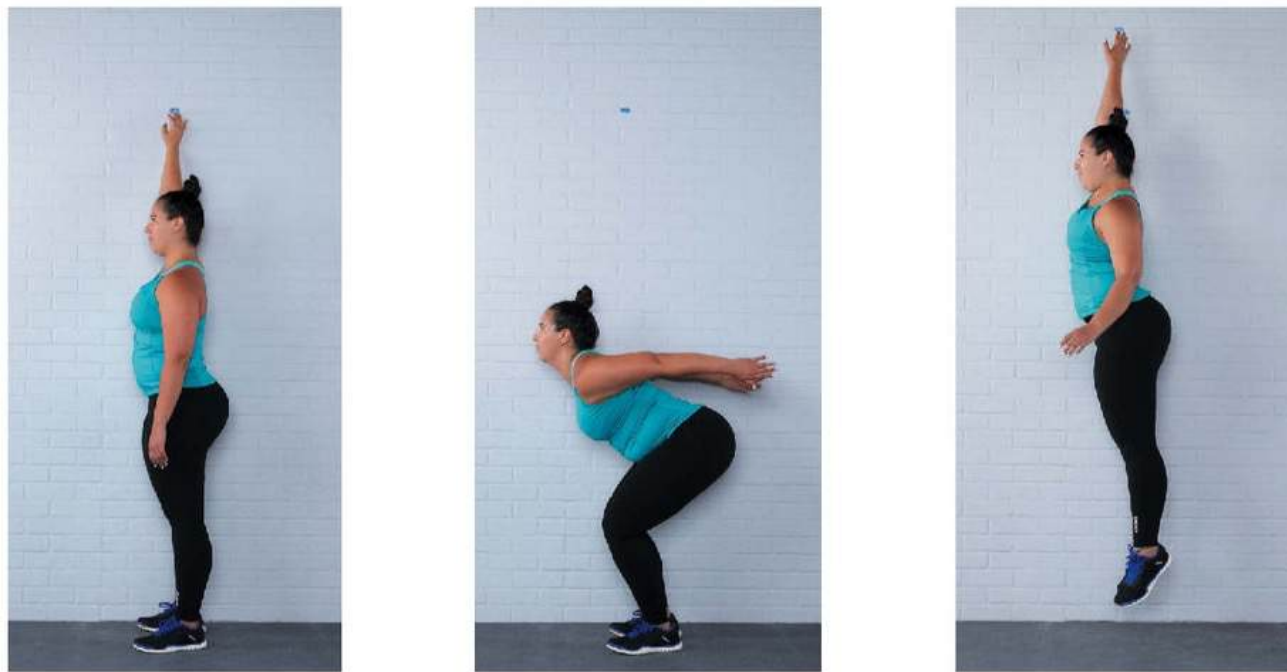


Figure 10-60: Vertical jump assessment

VIDEO

- <https://www.youtube.com/watch?v=zMQR6T2HVV0>
- Focus on 1:30 to 2:30 which shows how to perform the vertical jump test
- For the purpose of consistency with the book (previous slide), we will notice the vertex stand for measurement of vertical jump height
- Note how the jump is executed
- Go through the pointers highlighted in the previous slide
- Items required: wall, chalk and measuring tape (if this is not possible, then use vertex, but have to let students know that this is a variation of what is taught in the book)

VERTICAL JUMP ASSESSMENT

Men						
Age (years)	15-19	20-29	30-39	40-49	50-59	60-69
Excellent	≥56	≥58	≥52	≥43	≥41	≥33
Very good	51-55	54-57	46-51	36-42	34-40	29-32
Good	46-50	48-53	40-45	32-35	28-33	25-28
Fair	42-45	42-47	31-39	26-31	18-27	18-24
Needs Improvement	≤41	≤41	≤30	≤25	≤17	≤17

Women						
Age (years)	15-19	20-29	30-39	40-49	50-59	60-69
Excellent	≥40	≥38	≥36	≥31	≥25	≥19
Very good	36-39	34-37	32-35	27-30	21-24	15-18
Good	32-35	29-33	28-31	23-26	16-20	11-14
Fair	28-31	25-28	24-27	18-22	10-15	7-10
Needs improvement	≤27	≤24	≤23	≤17	≤9	≤6

Table 10-28: Norms for the Vertical Jump Assessment (cm)

SPEED, AGILITY, AND QUICKNESS ASSESSMENT

- Maximal effort and swift limb movement
- Adequate warm-up is required
 - Light jog/cardiorespiratory activity combined with short sprints
 - Light, dynamic stretching of the lower limbs (quadriceps, hamstrings, calves, and hip flexors)



T-TEST

- Objective: Assess multidirectional movement ability

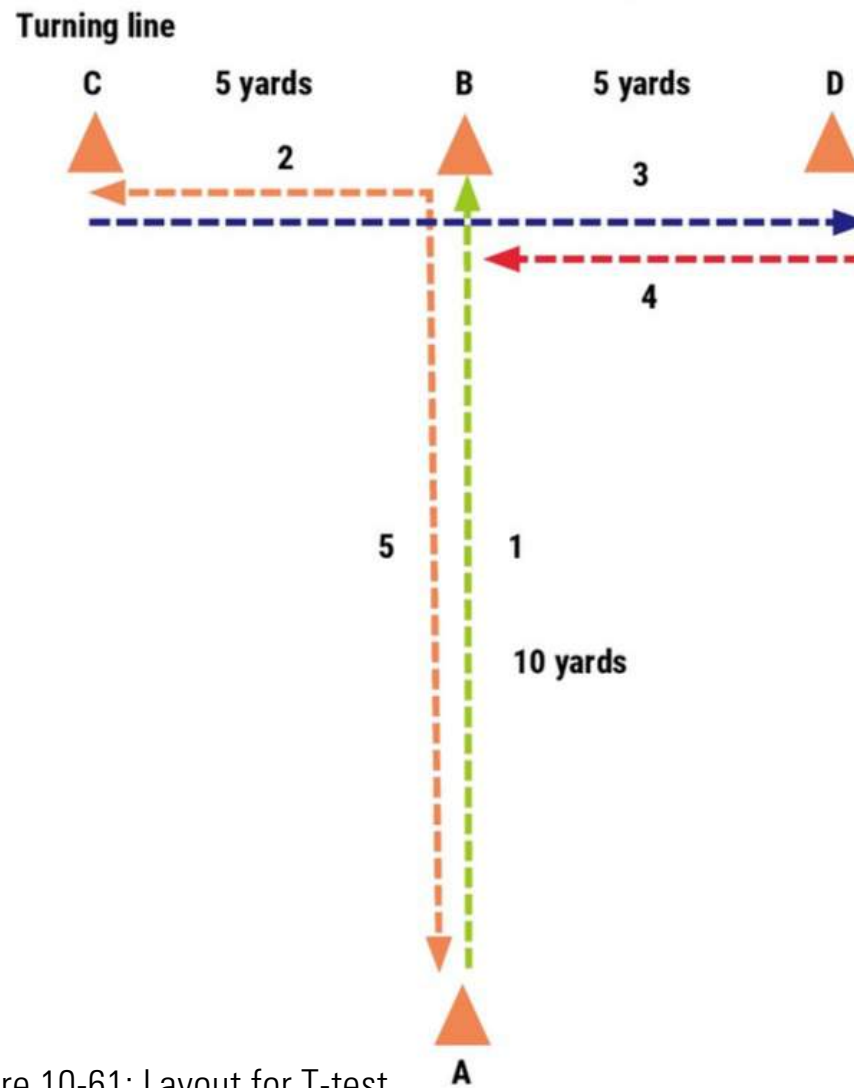


Figure 10-61: Layout for T-test

VIDEO

- https://www.youtube.com/watch?v=1UOP7h0eH_8
- The video shows how the T test is executed
- Note the steps which the performer has to take to complete the test (e.g, which hands to touch the cone, how to move around, where to face, etc)
- Go through the pointers highlighted in the previous slide
- Items required: cones, timer